

Name: Manahil Habib

Section: 5-B

Reg no: 2023-BSE-042

Lab 7

Task 1

1. Print all environment variables:

```
manahil@ubuntu:~$ printenv

SHELL=/bin/bash
CREDENTIALS_DIRECTORY=/run/credentials/getty@tty1.service
MEMORY_PRESSURE_WRITE=c29tZSAYMDAwMDAgMjAwMDAwMAA=
XDG_SEAT=seat0
PWD=/home/manahil
LOGNAME=manahil
XDG_SESSION_TYPE=tty
SYSTEMD_EXEC_PID=1394
HOME=/home/manahil
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33:01:cd=40;33:01:or=40;31:01:mi=00:su=37;41:sg=30;43:ca=00:tw=30
x=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lzh=01;31:*.lzm=01;31:*.tlz=01;31:*.txz=01
1;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzt=01;31:*.bz2=01;31:*.bz
2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;3
31:*.r2=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;3
35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:
:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:
*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=0
01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;3
36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.~*=00;90:*.#*=00;90:*.bak=00;90:*.crdownload=00;90:
g-new=00;90:*.dpg-old=00;90:*.dpg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=
=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
MEMORY_PRESSURE_WATCH=sys/fs/cgroup/system.slice/system-getty.slice/getty@tty1.service/memory.pressure
INVOCATION_ID=8668bad1b1214e4eb9917f1af343042c
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=linux
LESSOPEN=| /usr/bin/lesspipe %s
USER=manahil
SHLVL=1
XDG_VTNR=1
XDG_SESSION_ID=1
XDG_RUNTIME_DIR=/run/user/1000
XDG_DATA_DIRS=/usr/share/gnome:/usr/local/share:/usr/share:/var/lib/napd/desktop
HUSHLOGIN=FALSE
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
MAIL=/var/mail/manahil
_=/usr/bin/printenv
manahil@ubuntu:~$
```

2. Filter for SHELL, HOME and USER — run these greps together and capture one combined screenshot:

```
manahil@ubuntu:~$ 
manahil@ubuntu:~$ printenv | grep SHELL
SHELL=/bin/bash
manahil@ubuntu:~$ printenv | grep HOME
HOME=/home/manahil
manahil@ubuntu:~$ printenv | grep USER
USER=manahil
manahil@ubuntu:~$
```

Task 2

1. Define all DB_* variables (run the three exports one after another). Capture them in one screenshot showing the three export commands and their execution:

```
export DB_URL="postgres://db.example.local:5432/mydb"
```

```
USER=manahil
manahil@ubuntu:~$ export DB_URL="postgres://db.example.local:5432/mydb"
manahil@ubuntu:~$ export DB_USER="labuser"
manahil@ubuntu:~$ export DB_PASSWORD="labpass123"
manahil@ubuntu:~$ _
```

2. Echo the three variables (run the three echo commands together) and capture one screenshot showing their outputs:

```
manahil@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
manahil@ubuntu:~$ echo "$DB_USER"
labuser
manahil@ubuntu:~$ echo "$DB_PASSWORD"
labpass123
manahil@ubuntu:~$ _
```

3. Show all DB_ variables with a single grep command (capture that output):

```
labpass123
manahil@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
manahil@ubuntu:~$
```

4. Close the bash session (e.g., exit) and reopen a new terminal. Verify the variables are gone by running the echo(s) and the grep together; capture both checks in one screenshot:

```
manahil@ubuntu:~$ echo "$DB_URL"
manahil@ubuntu:~$ printenv | grep '^DB_'
manahil@ubuntu:~$
```

Task 3

1. Open ~/.bashrc in an editor and append the three export lines. Capture the editor showing the three lines added (single screenshot):

```

# Lab 7 persistent DB variables
#
export DB_URL="postgres://db.example.local:5432/mydb"
#
export DB_USER="labuser"
#
export DB_PASSWORD="labpass123"
#
fi

```

2. Source ~/.bashrc and capture the source command in one screenshot together with the next verification commands (grouped): run source ~/.bashrc and then immediately run the three echoes and a single grep, capturing all of these in one screenshot:

```

manahil@ubuntu:~$ source ~/.bashrc
manahil@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
manahil@ubuntu:~$ echo "$DB_USER"
labuser
manahil@ubuntu:~$ echo "$DB_PASSWORD"
labpass123
manahil@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
manahil@ubuntu:~$

```

3. Close and reopen terminal. Verify persistence by running one echo and the grep together — capture both in one screenshot:

```

manahil@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
manahil@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
manahil@ubuntu:~$

```

Task 4

1. View /etc/environment:

```

manahil@ubuntu:~$ sudo cat /etc/environment

PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
manahil@ubuntu:~$ _

```

2. Show current PATH (single print) — this is not a grep, but keep it as a dedicated screenshot:

```
manahil@ubuntu:~$ echo "$PATH"
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
manahil@ubuntu:~$
manahil@ubuntu:~$ _
```

3. Edit /etc/environment and add Class:

```
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-<your_class_name>"
```

```
manahil@ubuntu:~$ sudo cat /etc/environment

PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-<your_class_name>"

manahil@ubuntu:~$
```

4. Re-login or open a new shell and show Class and PATH together (grouped prints):
run echo \$Class and echo \$PATH together and capture in a single screenshot:

```
manahil@ubuntu:~$ echo $Class
CC-<your_class_name>
manahil@ubuntu:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
manahil@ubuntu:~$
```

5. Create welcome script on Desktop and make it executable (capture the heredoc creation and chmod together in one screenshot if possible):

```
manahil@ubuntu:~$ cat > ~/Desktop/welcome <<'EOF'
>
> #!/bin/bash
>
> echo "Welcome to Cloud Computing $USER"
>
> EOF

manahil@ubuntu:~$
manahil@ubuntu:~$ chmod +x ~/Desktop/welcome
manahil@ubuntu:~$ ls -l ~/Desktop/welcome
-rwxrwxr-x 1 manahil manahil 55 Nov  6 04:32 /home/manahil/Desktop/welcome
manahil@ubuntu:~$
manahil@ubuntu:~$ _
```

6. Run the script from Desktop using ./welcome (dedicated screenshot):

```
manahil@ubuntu:~$ cd ~/Desktop
manahil@ubuntu:~/Desktop$ ./welcome
Welcome to Cloud Computing manahil
manahil@ubuntu:~/Desktop$
```

7. Add Desktop to PATH in ~/.bashrc (show the editor containing the PATH addition in one screenshot) and then source; capture the source and a single invocation of welcome (from home) together:

```
PATH=$PATH:~/Desktop
```

```
export PATH
```

```
"~/ .bashrc" 128L, 3958B written
```

```
manahil@ubuntu:~/Desktop$ source ~/.bashrc
manahil@ubuntu:~/Desktop$ cd ~
manahil@ubuntu:~$ welcome
Welcome to Cloud Computing manahil
manahil@ubuntu:~$
manahil@ubuntu:~$
```

Task 5

1. Enable ufw and show status (group both commands in one screenshot if you run them together):

```
manahil@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
3389/tcp ALLOW IN Anywhere
3389/tcp (v6) ALLOW IN Anywhere (v6)

manahil@ubuntu:~$
```

2. Deny TCP port 22 and show status (run deny and status numbered together and capture in one screenshot):

```
manahil@ubuntu:~$ sudo ufw deny proto tcp to any port 22
Rule added
Rule added (v6)
manahil@ubuntu:~$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 3389/tcp ALLOW IN Anywhere
[ 2] 22/tcp DENY IN Anywhere
[ 3] 3389/tcp (v6) ALLOW IN Anywhere (v6)
[ 4] 22/tcp (v6) DENY IN Anywhere (v6)

manahil@ubuntu:~$
```

3. From Windows host attempt to SSH (expected to fail) — capture the host-side SSH attempt in one screenshot:

```
ssh: connect to host 192.168.58.139 port 22: Connection timed out
PS C:\Users\HABIB TARIQ>
```

4. Allow SSH back and reload, then show status (group allow, reload, status in one screenshot if run together):

```
manahil@ubuntu:~$ sudo ufw allow proto tcp to any port 22
Rule updated
Rule updated (v6)
manahil@ubuntu:~$ sudo ufw reload
Firewall reloaded
manahil@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
3389/tcp ALLOW Anywhere
22/tcp ALLOW Anywhere
3389/tcp (v6) ALLOW Anywhere (v6)
22/tcp (v6) ALLOW Anywhere (v6)

manahil@ubuntu:~$ _
```

5. From Windows host attempt SSH again (should succeed) — capture successful login in one screenshot:

```

PS C:\Users\HABIB TARIQ> ssh manahil@192.168.58.139
manahil@192.168.58.139's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Nov  6 06:21:35 AM UTC 2025

System load:  0.87               Processes:            237
Usage of /:   95.0% of 9.75GB    Users logged in:     1
Memory usage: 21%               IPv4 address for ens33: 192.168.58.139
Swap usage:   0%

=> / is using 95.0% of 9.75GB

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

Last login: Fri Oct 24 04:39:09 2025 from 192.168.58.1
manahil@ubuntu:~$

```

Task 6

1. Generate ed25519 key pair (if needed) and show the generated files in one screenshot (run ssh-keygen and then list ~/.ssh):

```
ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -C "lab_key"
```

```

manahil@ubuntu:~$ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -C "lab_key"
Generating public/private ed25519 key pair.
/home/manahil/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Saving key "/home/manahil/.ssh/id_ed25519" failed: No space left on device
manahil@ubuntu:~$

```

ls -la ~/.ssh

```
manahil@ubuntu:~$ ls -la ~/.ssh
total 40
drwx----- 2 manahil manahil 4096 Nov  6 06:24 .
drwxr-x--- 27 manahil manahil 4096 Nov  6 04:35 ..
-rw----- 1 manahil manahil    0 Sep 26 07:25 authorized_keys
-rw-rw-r-- 1 manahil manahil   96 Sep 27 09:00 config
-rw----- 1 manahil manahil  419 Sep 27 12:29 id_ed25519.github
-rw----- 1 manahil manahil  419 Sep 27 08:59 id_ed25519.github
-rw-r--r-- 1 manahil manahil  109 Sep 27 12:29 id_ed25519.github.pub
-rw-r--r-- 1 manahil manahil  109 Sep 27 08:59 id_ed25519.github.pub
-rw-r--r-- 1 manahil manahil  109 Sep 26 14:23 id_ed25519.pub
-rw----- 1 manahil manahil 1120 Sep 28 14:18 known_hosts
-rw-r--r-- 1 manahil manahil  284 Sep 28 14:18 known_hosts.old
manahil@ubuntu:~$
```

2. Show the public key content (single screenshot):

type %env:USERPROFILE\.ssh\id_ed25519.pub

```
PS C:\Users\HABIB TARIQ> type %env:USERPROFILE\.ssh\id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICR2om7wF4Qy3Y3lJOIBQ56B6zTfL8JIHGSiHF2XXHnV 22-22411-042@se.fjwu.edu.pk
PS C:\Users\HABIB TARIQ>
```

or on Git Bash: cat ~/.ssh/id_ed25519.pub

```
manahil@ubuntu:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHCACr59bwaJQaXH0dLJe6D0Wo/dfJzKzfGvVKvq06D 23-22411-042@se.fjwu.edu.pk
manahil@ubuntu:~$
```

3. (Optional) Show known_hosts after first connection (single screenshot):

type %env:USERPROFILE\.ssh\known_hosts

```
PS C:\Users\HABIB TARIQ> type %env:USERPROFILE\.ssh\known_hosts
192.168.58.139 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIL8pGJwHHqC8/ZpIh+DjZ+MHkIpAef4igP5ORAPMWJdu
192.168.58.139 ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGCiownr8/0EUZ9xgdT20tWvpKu93RD/6pRi44IvQPZ4LLjFwiiEwZxWJEzqsFC6hB9YA
dfV0YagbSTx2vWYvCJL11FUqyG3SHAL+sT0LPTMOKt1akmMbDeZyy4Mc2dKlWEBHSYj0mECJ03A1P+00dLHKufBzcj+0p5GhMvTyBXHOKTZ4GPgdNeFOEOHP
IF+sr+Kz0G/sTtd5B3ZqkLpL4G9tiVv2jXiFGgWxoBqS8ojUZLVdGWYa9pYmqiLjiyAtgSSknJbNLZYoiS8GMYcdcq+QteYv0STJTp2bd5cTy1qkd0a+2mp
Bqt0/Lkd91MgkHEgix5zytjSfmbEjq3R7LcOxbDLWPeRgSeKCa++uHxRc1L/uE10B/916UJvL7Ls1VfLkmV2w0FoyW+jaq6yCS/L5IQ3i082VfAfbnj7FPA
AMzFME25pBtNsEwJmgh0LLpplmQzpiCvvtY7GthXupkZ1DjPVHTdfFWp07Ss/KJ4lwM5NsT3982ugumdyBj5n0=
192.168.58.139 ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlldHAYNTYAAAAIbmlldHAYNTYAAABBBBcZjBTgrwF65+pJ7Js4z3GWtkYjCCNcK
Vsqd6R5sH57RZhXGa6GQT1odp3j+zfaTUzqa6wtKvMbaxVj2yv0X4A=
PS C:\Users\HABIB TARIQ>
```

B. On Ubuntu server — group related server-side commands:

1. Inspect .ssh and authorized_keys before adding key (capture both outputs in one screenshot):

```
manahil@ubuntu:~$ ls -la ~/.ssh
total 40
drwx----- 2 manahil manahil 4096 Nov  6 06:24 .
drwxr-x--- 27 manahil manahil 4096 Nov  6 04:35 ..
-rw----- 1 manahil manahil    0 Sep 26 07:25 authorized_keys
-rw-rw-r-- 1 manahil manahil   96 Sep 27 09:00 config
-rw----- 1 manahil manahil  419 Sep 27 12:29 id_ed25519.github
-rw----- 1 manahil manahil  419 Sep 27 08:59 id_ed25519.github
-rw-r--r-- 1 manahil manahil  109 Sep 27 12:29 id_ed25519.github.pub
-rw-r--r-- 1 manahil manahil  109 Sep 27 08:59 id_ed25519.github.pub
-rw-r--r-- 1 manahil manahil  109 Sep 26 14:23 id_ed25519.pub
-rw----- 1 manahil manahil 1120 Sep 28 14:18 known_hosts
-rw-r--r-- 1 manahil manahil  284 Sep 28 14:18 known_hosts.old
manahil@ubuntu:~$ cat ~/.ssh/authorized_keys | true
manahil@ubuntu:~$ _
```

2. Create ~/.ssh, set permissions, append the public key (manual method), and show the resulting authorized_keys — capture the mkdir/chmod/echo and the final cat in one screenshot:

```
manahil@ubuntu:~$ mkdir -p ~/.ssh
manahil@ubuntu:~$ chmod 700 ~/.ssh
manahil@ubuntu:~$ echo "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI... lab_key" >> ~/.ssh/authorized_keys
manahil@ubuntu:~$ chmod 600 ~/.ssh/authorized_keys
manahil@ubuntu:~$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI... lab_key
manahil@ubuntu:~$
```

3. From Windows host test passwordless login (capture successful login in one screenshot):

ssh username@<server_ip>

```
PS C:\Users\HABIB TARIQ> ssh manahil@192.168.58.139
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Nov  6 08:43:20 AM UTC 2025

System load:  0.01               Processes:            239
Usage of /:   94.6% of 9.75GB    Users logged in:     1
Memory usage: 21%               IPv4 address for ens33: 192.168.58.1
Swap usage:   0%

=> / is using 94.6% of 9.75GB

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Nov  6 08:15:09 2025 from 192.168.58.1
manahil@ubuntu:~$
```

4. Also demonstrate explicit identity usage (single screenshot):

`ssh -i ~/.ssh/id_ed25519 username@<server_ip>`

```
PS C:\Users\HABIB TARIQ> ssh -vvv -i ~/.ssh/id_ed25519 manahil@192.168.58.139
OpenSSH_for_Windows_9.5p2, LibreSSL 3.8.2
debug3: Failed to open file:C:/Users/HABIB TARIQ/.ssh/config error:2
debug3: Failed to open file:C:/ProgramData/ssh/ssh_config error:2
debug2: resolve_canonicalize: hostname 192.168.58.139 is address
debug3: expanded UserKnownHostsFile '~/.ssh/known_hosts' -> 'C:\\Users\\HABIB TARIQ/.ssh/known_hosts'
debug3: expanded UserKnownHostsFile '~/.ssh/known_hosts2' -> 'C:\\Users\\HABIB TARIQ/.ssh/known_hosts2'
debug3: ssh_connect_direct: entering
debug1: Connecting to 192.168.58.139 [192.168.58.139] port 22.
debug1: Connection established.
debug1: identity file C:\\Users\\HABIB TARIQ/.ssh/id_ed25519 type 3
debug3: Failed to open file:C:/Users/HABIB TARIQ/.ssh/id_ed25519-cert error:2
debug3: Failed to open file:C:/Users/HABIB TARIQ/.ssh/id_ed25519-cert.pub error:2
debug3: failed to open file:C:/Users/HABIB TARIQ/.ssh/id_ed25519-cert error:2
debug1: identity file C:\\Users\\HABIB TARIQ/.ssh/id_ed25519-cert type -1
debug1: Local version string SSH-2.0-OpenSSH_for_Windows_9.5
debug1: Remote protocol version 2.0, remote software version OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
debug1: compat_banner: match: OpenSSH_9.6p1 Ubuntu-3ubuntu13.14 pat OpenSSH* compat 0x04000000
debug2: fd 3 setting O_NONBLOCK
debug1: Authenticating to 192.168.58.139:22 as 'manahil'
debug3: record_hostkey: found key type ED25519 in file C:\\Users\\HABIB TARIQ/.ssh/known_hosts:1
debug3: record_hostkey: found key type RSA in file C:\\Users\\HABIB TARIQ/.ssh/known_hosts:2
debug3: record_hostkey: found key type ECDSA in file C:\\Users\\HABIB TARIQ/.ssh/known_hosts:3
debug3: load_hostkeys_file: loaded 3 keys from 192.168.58.139
debug3: Failed to open file:C:/Users/HABIB TARIQ/.ssh/known_hosts2 error:2
debug1: load_hostkeys: fopen C:\\Users\\HABIB TARIQ/.ssh/known_hosts2: No such file or directory
debug3: Failed to open file:C:/ProgramData/ssh/ssh_known_hosts error:2
debug1: load_hostkeys: fopen _PROGRAMDATA_\\ssh/ssh_known_hosts: No such file or directory
```

System information as of Thu Nov 6 08:44:47 AM UTC 2025

System load:	0.08	Processes:	246
Usage of /:	94.6% of 9.75GB	Users logged in:	1
Memory usage:	22%	IPv4 address for ens33:	192.168.58.139
Swap usage:	0%		

=> / is using 94.6% of 9.75GB

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.

10 of these updates are standard security updates.

To see these additional updates run: `apt list --upgradable`

12 additional security updates can be applied with ESM Apps.

Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

Last login: Thu Nov 6 08:44:12 2025 from 192.168.58.1

manahil@ubuntu:~\$
